Accelerated Math II/III
Irrational & Complex Root Theorems

Names: _____
Period: _____

Complete #1-37 odd.

A polynomial equation with rational coefficients has the given roots. Find two additional roots.

1. $2 + 3i$ and $\sqrt{7}$
2. $3 - \sqrt{2}$ and $1 + \sqrt{3}$
3. $-4i$ and $6 - i$
4. $5 - \sqrt{6}$ and $-2 + \sqrt{10}$

Find a fourth-degree polynomial equation with integer coefficients that has the given numbers as roots.

5. $2i$ and $4 - i$
6. $\sqrt{2}$ and $2 - \sqrt{3}$
7. $3i$ and $\sqrt{6}$
8. $2 + i$ and $1 - \sqrt{5}$

Find the roots of each polynomial equation.

9. $x^3 - 5x^2 + 2x + 8 = 0$
10. $x^3 + x^2 - 17x + 15 = 0$
11. $2x^3 + 13x^2 + 17x - 12 = 0$
12. $x^3 - x^2 - 34x - 56 = 0$
13. $x^3 - 18x + 27 = 0$
14. $x^4 - 5x^2 + 4 = 0$
15. $x^3 - 6x^2 + 13x - 10 = 0$
16. $x^3 - 5x^2 + 4x + 10 = 0$
17. $x^3 - 5x^2 + 17x - 13 = 0$
18. $x^3 + x + 10 = 0$
19. $x^3 - 5x^2 - x + 5 = 0$
20. $x^3 - 12x + 16 = 0$
21. $x^3 - 2x^2 - 5x + 6 = 0$
22. $x^3 - 8x^2 - 200 = 0$
23. $x^3 + x^2 - 5x + 3 = 0$
24. $4x^3 - 12x^2 - x + 3 = 0$
25. $x^3 + x^2 - 7x + 2 = 0$
26. $12x^3 + 31x^2 - 17x - 6 = 0$

Use the Rational Root Theorem to list all possible rational roots for each polynomial equation. Then find any actual rational roots.

27. $x^3 + 5x^2 - 2x - 15 = 0$
28. $36x^3 + 144x^2 - x - 4 = 0$
29. $2x^3 + 5x^2 + 4x + 1 = 0$
30. $12x^4 + 14x^3 - 5x^2 - 14x - 4 = 0$
31. $5x^3 - 11x^2 + 7x - 1 = 0$
32. $x^3 + 81x^2 - 49x - 49 = 0$

Find a third-degree polynomial equation with rational coefficients that has the given numbers as roots.

33. $3, 2 - i$
34. $5, 2i$
35. $-1, 3 + i$
36. $-7, i$
37. $-4, 4i$
38. $6, 3 - 2i$

Answers

1. $2 - 3i, -\sqrt{7}$ **2.** $3 + \sqrt{2}, 1 - \sqrt{3}$ **3.** $4i, 6 + i$
4. $5 + \sqrt{6}, -2 - \sqrt{10}$ **5.** $x^4 - 8x^3 + 21x^2 - 32x + 68$
6. $x^4 - 4x^3 - x^2 + 8x - 2$ **7.** $x^4 + 3x^2 - 54$
8. $x^4 - 6x^3 + 9x^2 + 6x - 20$ **9.** $4, 2, -1$ **10.** $3, 1, -5$
11. $-4, -3, \frac{1}{2}$ **12.** $7, -2, -4$ **13.** $3; \frac{-3 \pm 3\sqrt{5}}{2}$
14. $-2, -1, 1, 2$ **15.** $2, 2 \pm i$ **16.** $-1, 3 \pm i$ **17.** $1, 2 \pm 3i$
18. $-2, 1 \pm 2i$ **19.** $1, -1, 5$ **20.** $-4, 2$ **21.** $-2, 1, 3$
22. $10, -1 \pm i\sqrt{19}$ **23.** $1, -3$ **24.** $3, \frac{1}{2}, -\frac{1}{2}$
25. $2, \frac{-3 \pm \sqrt{13}}{2}$ **26.** $-3, \frac{2}{3}, -\frac{1}{4}$
27. $\pm 1, \pm 3, \pm 5, \pm 15$; none **28.** $\pm 1, \pm 2, \pm 4, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}, \pm\frac{1}{4}, \pm\frac{1}{6}, \pm\frac{1}{9}, \pm\frac{2}{9}, \pm\frac{4}{9}, \pm\frac{1}{12}, \pm\frac{1}{18}, \pm\frac{1}{36}; -4, -\frac{1}{6}, \frac{1}{6}$
29. $\pm 1, \pm\frac{1}{2}; -1, -\frac{1}{2}$ **30.** $\pm 1, \pm 2, \pm 4, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}, \pm\frac{1}{4}, \pm\frac{1}{6}, \pm\frac{1}{12}$; none **31.** $\pm 1, \pm\frac{1}{5}; \frac{1}{5}; 1$ **32.** $\pm 1, \pm 7, \pm 49$; none
33. $x^3 - 7x^2 + 17x - 15 = 0$ **34.** $x^3 - 5x^2 + 4x - 20 = 0$
35. $x^3 - 5x^2 + 4x + 10 = 0$ **36.** $x^3 + 7x^2 + x + 7 = 0$
37. $x^3 + 4x^2 + 16x + 64 = 0$ **38.** $x^3 - 12x^2 + 49x - 78 = 0$